



Area of a triangle

AREA IS THE COMPLETE SPACE INSIDE A TRIANGLE.

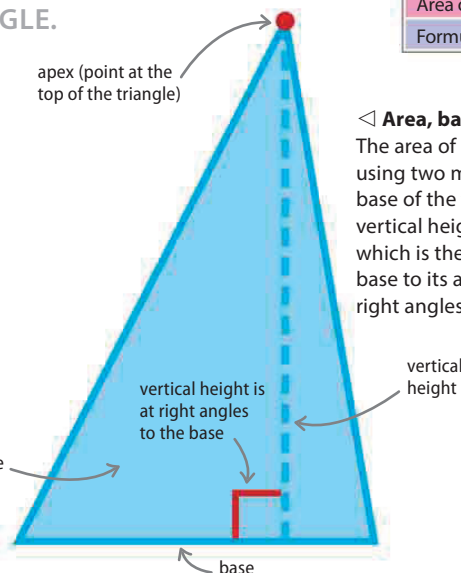
What is area?

The area of a shape is the amount of space that fits inside its outline, or perimeter. It is measured in squared units, such as cm^2 . If the length of the base and vertical height of a triangle are known, these values can be used to find the area of the triangle, using a simple formula, which is shown below.

$$\text{area} = \frac{1}{2} \times \text{base} \times \text{vertical height}$$

this is the formula for finding the area of a triangle

area is the space inside a triangle's frame



SEE ALSO

◀ 116–117 Triangles

Area of a circle 142–143 ▶

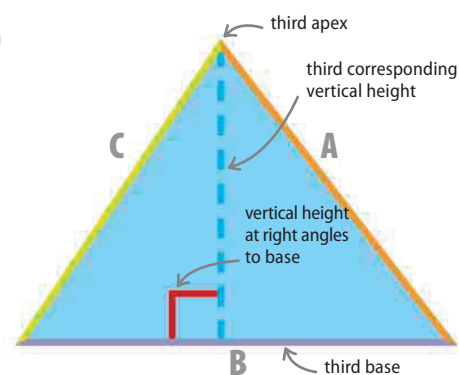
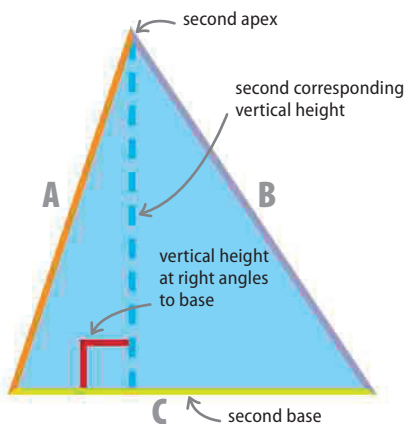
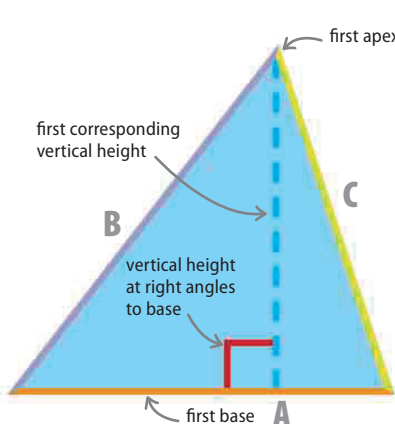
Formulas 177–179 ▶

Area, base, and height

The area of a triangle is found using two measurements: the base of the triangle and the vertical height of the triangle, which is the distance from its base to its apex, measured at right angles to the base.

Base and vertical height

Finding the area of a triangle requires two measurements, the base and the vertical height. The side on which a triangle “sits” is called the base. The vertical height is a line formed at right angles to the base from the apex. Any one of the three sides of a triangle can act as the base in the area formula.



△ First base

The area of the triangle can be found using the orange side (A) as the “base” needed for the formula. The corresponding vertical height is the distance from the base of the triangle to its apex (highest point).



△ Second base

Any one of the triangle's three sides can act as its base. Here the triangle is rotated so that the green side (C) is its base. The corresponding vertical height is the distance from the base to the apex.



△ Third base

The triangle is rotated again, so that the purple side (B) is its base. The corresponding vertical height is the distance from the base to the apex. The area of the triangle is the same, whichever side is used as the base in the formula.